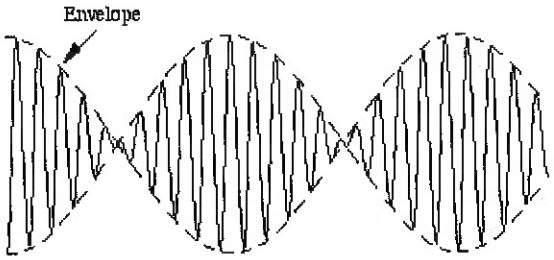
Electromagnetic Fields 2019 – Solutions

1a) In complex notation, the amplitude variation of a single wave travelling in the z-direction at angular frequency ω can be written as $A(z, t) = A_0 \exp\{j\omega t - kz)\}$. The phase velocity $v_{ph} = \omega/k = f\lambda$ gives the speed of the wave.

[2]

A single wave cannot carry any information. To send data, a carrier must be modulated. The envelope (which contains the information) then travels at a slightly different speed. Consider the simplest possible AM signal, formed from two signals of angular frequencies $\omega + d\omega$ and $\omega - d\omega$. The corresponding k-values are $k + dk$ and $k - dk$. For equal amplitudes, the combined signal is $A(z, t) = A_0 [\exp\{j((\omega + d\omega)t - (k + dk)z)\} + \exp\{j((\omega - d\omega)t - (k - dk)z)\}]$.
This can be written alternatively as $A = 2A_0 \exp\{j\omega t - kz)\} \cos\{d\omega t - dk z\}$. The speed of the carrier is $v_p = \omega/k$ as before. However, the speed of the envelope is the group velocity $v_g = d\omega/dk$.

[4]



[2]

b) Absorption, scattering and diffraction are the three main mechanisms limiting line of sight communications. They effect different frequency ranges in different ways:

[2]

- Absorption - Excitation of electronic transitions in the molecules of the earth's atmosphere at visible/UV wavelengths, and molecular vibrational transitions at infrared wavelengths. Losses are concentrated in absorption bands.

[2]

- Scattering - Rayleigh scattering arises from inhomogeneities (e.g. soot particles) and small-scale fluctuations in the molecular arrangement of the atmosphere. Scattering losses rise rapidly at short wavelengths (rising as $1/\lambda^4$).

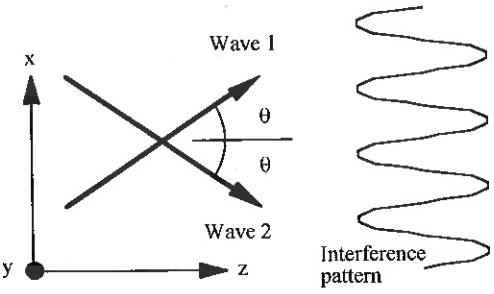
[2]

- Diffraction - Spreading of a bounded beam as it propagates. Diffraction increases rapidly as the beam dimensions approach that of the wavelength, and is avoided using guided waves.

[2]

c) Interference is the result of coherent addition of two or more waves of the same frequency. For example, if two waves cross at equal and opposite angles θ, we can obtain the total time-independent field by summing their individual contributions as:
$\underline{E} = j E_y \{\exp[-jk_0(z \cos(\theta) + x \sin(\theta))] + \exp[-jk_0(z \cos(\theta) - x \sin(\theta))]\}$
The combined field can be written alternatively as $\underline{E} = j\, 2E_y \exp[-jk_0 z \cos(\theta)] \cos[k_0 x \sin(\theta)]$
The irradiance depends on $|\underline{E}|^2$, and its magnitude must therefore vary as $\cos^2[(k_0 x \sin(\theta)]$
This is a sinusoidal interference pattern.

[4]



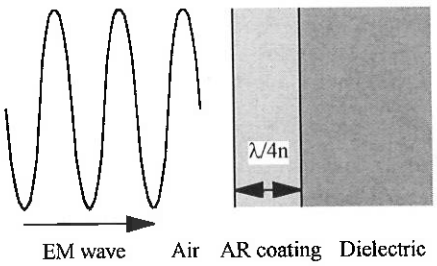
[2+2=4]

d) Anti-reflection coatings are used to reduce the reflectivity at interfaces, typically between glass and air. Operation is identical to a quarter wave transformer, and a quarter-wave thickness of material simply used as intermediate coating.

[2]

For glasses (for example), the design wavelength is $\lambda \sim 0.5$ μm (green light, the wavelength of peak eye sensitivity). The impedance of the coating is then chosen to be the geometric mean of the values for air and the dielectric used. Since the impedance of a general medium with index n is Z_0/n, the correct result can be achieved by taking the refractive index of the coating to be the geometric mean of the values for air and the dielectric used, so that $n_{coating} = (n_{air}n_{diel})^{1/2}$. For plastics ($n_{diel} \sim 1.6$), we need $n_{coat} \sim 1.3$. Magnesium fluoride (MgF) is then a suitable material.

[4]



[2]

e) Because all materials have a refractive index of approximately unity at high frequency, it is extremely difficult to form a lens for X-rays, and any conventional imaging system. A CAT (computer-aided tomography) scanner operates without the need for a lens.

[2]

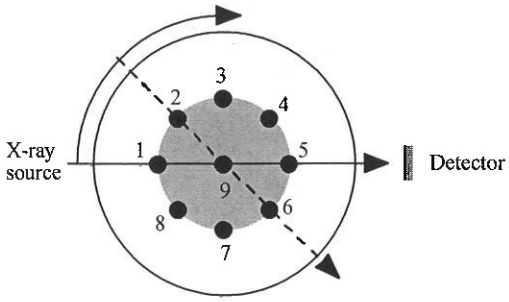
It consists of an X-ray source and a detector array, placed on either side of the patient. The source and detectors are rotated round the patient, and signals are measured at different angular positions. Each signal is determined by the absorption between source and detector. Simultaneous equations are then solved to find the local absorber values and hence the distribution of absorption (the X-ray image). The example below shows the principle. The 'patient' is modelled as nine absorbers, whose values we wish to find.

The measured absorption A_1 along the full line is $A_1 = a_1 + a_5 + a_9$

While the absorption along the dashed line is $A_2 = a_2 + a_6 + a_9$

These are two simultaneous equations. Rotating the source and detector together we can form eight equations, one fewer than is needed to solve for the unknown values a_1, a_2 … a_9. However additional equations can easily be provided using additional detectors. A matrix equation is then set up and inverted to find the unknown absorber values and build up a picture of the patient.

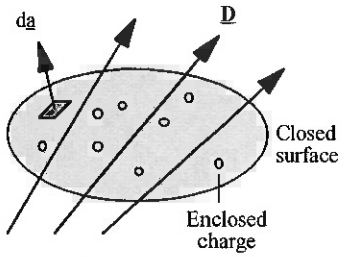
[4]



[2]

2a) Consider the electric flux out of a closed surface enclosing charges
In integral form, Gauss' Law states that $\iint_A \underline{D} \cdot d\underline{a} = \iiint_V \rho \, dv$
Here $d\underline{a}$ is a vector representing a small area of the surface
The modulus of $d\underline{a}$ is the size of the area
The direction of $d\underline{a}$ is the normal vector to the area
$\underline{D} \cdot d\underline{a}$ is therefore the amount of electric flux normal to the small area
$\iint_A \underline{D} \cdot d\underline{a}$ is the sum over the surface, i.e. the total flux out
If ρ is the charge density then $\iiint_V \rho \, dv$ is the integral of the charge over the volume
Consequently the integral equation also states that Flux out = charge enclosed

[2]

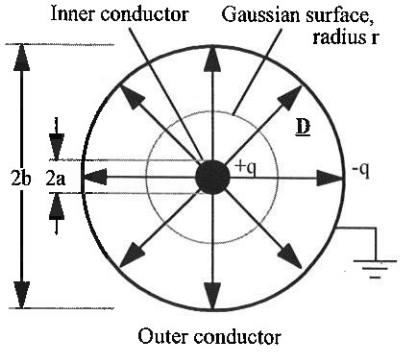


[2]

Gauss' law can be used to calculate the capacitance between two electrodes as follows. The charge distribution ±Q on the electrodes is first assumed. A Gaussian surface enclosing one set of charges is then defined, and Gauss' law is used to find the flux leaving the surface from the enclosed charge. The constitutive relation $\underline{D} = \varepsilon \underline{E}$ is then used to find the electric field from the flux density. The electric field is then integrated to find the voltage V between the electrodes. Finally, the capacitance C is found from the relation Q = CV.

[3]

b) Co-axial cables consist of an inner metal conductor of radius a surrounded by a layer of plastic insulation of dielectric constant ε, and a braided outer conductor of radius b that acts as a screen.



[1]

The appropriate Gaussian surface in this case is a cylinder of radius r.

[1]

To find the capacitance per unit length (pul), assume that there is a line charge +q per unit length on the inner conductor, and a matching charge –q on the outer conductor. The charge pul inside the Gaussian surface is therefore +q. These charges will create an electrical flux. The flux density $\underline{D}$ must be radial by symmetry, so $\underline{D} = D_r \underline{r}$. The flux out of the Gaussian surface is therefore $2\pi r D_r$ pul and Gauss' Law then gives $2\pi r D_r = q$

[2]

The constitutive relation $\underline{D} = \varepsilon \underline{E}$ implies that $2\pi r\varepsilon\, E_r = q$ so that $E_r = q/2\pi r\varepsilon$
The potential V between the conductors can be found by integrating the electric field, as:
$V(b) - V(a) = -\int_a^b (q/2\pi r\varepsilon)\, dr = -(q/2\pi\varepsilon) \log_e(b/a)$
If the outer conductor is grounded, $V(a) = (q/2\pi\varepsilon) \log_e(b/a)$

[2]

The capacitance per unit length can be found from the relation $q = C_p V$ as:
$C_p = q/V = (2\pi\varepsilon) \div \log_e(b/a)$ F/m

[1]

c) Assuming that $C_p = (2\pi\varepsilon) \div \log_e(b/a)$ F/m and $L_p = (\mu_0/2\pi) \log_e(b/a)$ H/m
with a = 0.5 mm, b = 2.5 mm, and $\varepsilon = \varepsilon_0\varepsilon_r$ with $\varepsilon_r = 2.25$, we get:

$C_p = (2\pi \times 8.85 \times 10^{-12} \times 2.25) \div \log_e(2.5/0.5)$ F/m = 77.7 pF/m
$L_p = (4\pi \times 10^{-7}/2\pi) \log_e(2.5/0.5)$ H/m = 321.9 nH/m

[2]

The characteristic impedance is then $Z_0 = \sqrt{(L_p/C_p)} = 64.4\ \Omega$
The phase velocity is then $v = 1/\sqrt{(L_pC_p)} = 2 \times 10^8$ m/s

[2]

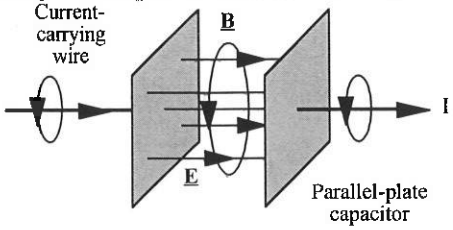
$v = c/n$, where $c = 3 \times 10^8$ m/s is the velocity of light in vacuo and $n = \sqrt{\varepsilon_r}$
Z_0 is real, since it is the property of an infinite line that will never return any injected power

[2]

3. a) In its original form (but with modern notation) Ampère's law would have been written as:
$\int_L \underline{\mathbf{H}} \cdot d\underline{L} = \iint_A \underline{\mathbf{J}} \cdot d\underline{a}$

[1]

Maxwell noticed that the existing theory could not really describe the continuity of current flow through a simple circuit containing wires connecting to a capacitor.



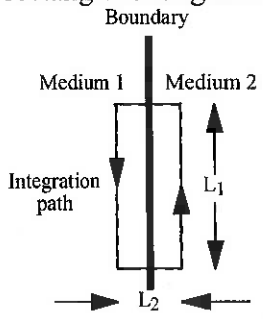
By Gauss' law, the electric field between plates is flux p.u.a = charge p.u.a. so $\varepsilon E = Q/A$
Maxwell noticed that $\varepsilon\, dE/dt = D/dt = I/A$ is really a current density. He called this new term $\underline{\mathbf{J}}_D$, the displacement current, and added it to Ampere's law, which then became:
$\int_L \underline{\mathbf{H}} \cdot d\underline{L} = \iint_A [\underline{\mathbf{J}} + \partial\underline{\mathbf{D}}/\partial t] \cdot d\underline{a}$

[2]

Here $\underline{\mathbf{J}}$ describes the flow of normal current (moving electrons) while $\underline{\mathbf{J}}_D = \partial\underline{\mathbf{D}}/\partial t$ describes the effect of time-varying fields. $\underline{\mathbf{J}}_D$ clearly increases as the rate of change of field (i.e. the frequency) rises. As a result, $\underline{\mathbf{J}}$ dominates at low frequency, while $\partial\underline{\mathbf{D}}/\partial t$ dominates at high frequency. It is then possible to have a flow of energy without any moving electrons at all. This unified the theory, and made the theoretical analysis of high frequency electromagnetic fields possible.

[2]

b) The figure below shows a boundary between two media. Medium 1 has a time-dependent magnetic field $\underline{\mathbf{H}}_1$, and Medium 2 has a corresponding field $\underline{\mathbf{H}}_2$. Define the integration path for the line integral as the rectangle shown. Two of the sides are parallel to the boundary and the rectangle is long and thin so that $L_2 << L_1$.



[2]

Ampere's law states that $\int_L \underline{\mathbf{H}} \cdot d\underline{L} = \iint_A [\underline{\mathbf{J}} + \partial\underline{\mathbf{D}}/\partial t] \cdot d\underline{a}$. If we let L_2 tend to zero, the area must tend to zero, so the surface integral must be zero. If L_1 is small but non-zero, the fields can be considered uniform. Integration then gives $\int_L \underline{\mathbf{H}} \cdot d\underline{L} = (\mathbf{H}_{t1} - \mathbf{H}_{t2})\, L_1$, where $\mathbf{H}_{t1}$ and $\mathbf{H}_{t2}$ are the components of the two fields parallel to the line elements. Consequently $\mathbf{H}_{t1} - \mathbf{H}_{t2} = 0$ and tangential components of $\underline{\mathbf{H}}$ must match across the boundary.

[3]

c) The power passing through 1 m^2 area is the power density S. For a plane wave (a suitable approximation to a spherical wave at this distance), $S = E_0^2/Z_0$, where E_0 is the peak electric field and $Z_0 = 377\ \Omega$ is the characteristic impedance of free space. Consequently the peak electric field is $E_0 = \sqrt{(SZ_0)}$. If S = 5 mW, $E_0 = (5 \times 10^{-3} \times 377)^{1/2} = 1.373$ V.

[3]

Assuming the antenna radiates isotropically, the power density S at a distance R is $P_T/4\pi r^2$, where P_T is the transmitter power. If $r = 10^3$ m and S = 5 mW, $P_T = 4\pi \times 10^6 \times 5 \times 10^{-3}$ W = 62.83 kW.

[2]

d) In general, the antenna gain is $G = \eta D$ where η is the antenna efficiency. However, if $\eta = 1$, G = D. Since the effective area is $A_e = \lambda^2 D/4\pi$, we obtain $A_e = \lambda^2 G/4\pi$ in this case. The wavelength is $\lambda = c/f$ where $c = 3 \times 10^8$ m/s is the velocity of light. If f = 10 GHz, $\lambda = 3 \times 10^8/10^{10} = 0.03$ m. If G = 5000 we then obtain $A_e = 0.03^2 \times 5000/4\pi = 0.358$ m^2.

[3]

Received power can be calculated from the radar equation $P_R = P_T(\eta_T^2 A_T^2 \sigma/4\pi r^4 \lambda^2)$.
Assuming $P_T = 62.83$ kW, an antenna with $\eta_T = 1$ and $A_T = 0.358$ m^2, a target cross section $\sigma =$ 10 m^2, a range $r = 10^4$ m and $\lambda = 0.03$ m we get:
$P_R = 62.83 \times 10^3 \times 0.358^2 \times 20 / (4\pi \times 10^{16} \times 0.03^2)$ W = 1.42 nW.

[2]